



Backint Certification Test Suite

1.50.2

Target Audience

- Development Partners

©2025 SAP SE or an SAP affiliate company. All rights reserved.

No part of this publication may be reproduced or transmitted in any form or for any purpose without the express permission of SAP SE or an SAP affiliate company.

The information contained herein may be changed without prior notice. Some software products marketed by SAP SE and its distributors contain proprietary software components of other software vendors. National product specifications may vary.

These materials are provided by SAP SE or an SAP affiliate company for informational purposes only, without representation or warranty of any kind, and SAP or its affiliated companies shall not be liable for errors or omissions with respect to the materials. The only warranties for SAP or SAP affiliate company products and services are those that are set forth in the express warranty statements accompanying such products and services, if any. Nothing herein should be construed as constituting an additional warranty.

SAP and other SAP products and services mentioned herein as well as their respective logos are trademarks or registered trademarks of SAP SE (or an SAP affiliate company) in Germany and other countries. All other product and service names mentioned are the trademarks of their respective companies. Please see <https://www.sap.com/about/legal/trademark.html> for additional trademark information and notices.

Contents

1 Document History	4
2 Introduction	5
2.1 Overview	5
3 Installation and Execution of the Backint Certification Test Suite	6
3.1 Installation	6
3.2 Execution	6
3.3 Test Catalog	8
3.3.1 Testcase #1: Recover Data	8
3.3.2 Testcase #2: Recover Database (point in time recovery)	8
3.3.3 Testcase #3: Inquire with wildcard #NULL	8
3.3.4 Testcase #4: Inquire single backup	8
3.3.5 Testcase #5: Inquire oldest two backups	8
3.3.6 Testcase #6: Inquire younger backup	8
3.3.7 Testcase #7: Inquire youngest and oldest backup	8
3.3.8 Testcase #8: Inquire youngest backup single ebid	9
3.3.9 Testcase #9: Delete youngest backup	9
3.3.10 Testcase #10: Inquire oldest backup single ebid	9
3.3.11 Testcase #11: Inquire deleted backup	9
3.3.12 Testcase #12: Inquire creation timestamp	9
3.4 Test Report	9

1 Document History

1.00

- Initial version

1.07

- migrated document source from wiki to LaTeX

1.50

- Extensions for Backint 1.5
- New parameter : `--backintVersion`
- New test case : Inquire creation time

1.50.1

- Update formatting
- Clarified SAP HANA versions needed

1.50.2

- Adapt installation instructions to shipment in HANA server package

2 Introduction

2.1 Overview

This document describes the *Backint Certification Test Suite*. This test suite includes several scenarios which require a multi-container SAP HANA installation with a tenant database which has the same name as SID.

The certification suite for Backint 1.5 and later requires an installation of HANA SPS05 or later.

3 Installation and Execution of the Backint Certification Test Suite

The following steps have to be executed as user System Administrator (sidadm).

3.1 Installation

The certification suite is shipped together with the server package of SAP HANA. When installing the server package using `hdb1cm` you will see the package being installed:

```
...  
Preparing package 'Backint Validation Tools'...  
...  
Installing package 'Backint Validation Tools'...  
...
```

The installed package is located in the directory `/usr/sap/<SID>/exe/hdb/sdk/backint` and is called `hdbbackintcertification_pack.tar.gz`. It is recommended that the archive `hdbbackintcertification_pack.tar.gz` is extracted into a subdirectory of the work directory `$DIR_INSTANCE/work` and extracted in a subdirectory. The extraction can be done using the `tar` command as follows:

```
tar -xvfz /usr/sap/<SID>/exe/hdb/sdk/backint/hdbbackintcertification_pack.tar.gz
```

If you have not chosen the default passwords for the database user `SYSTEM` and System Administrator (sidadm), change them accordingly, i.e.

user = `SYSTEM` and password = `manager`

user = `<sidadm>` and password = `trextrex`

Otherwise the certification suite will not run properly.

3.2 Execution

Move to the directory you have extracted the tools and execute the following statement:

```
python test_backup_backint_certification.py
```

This executes the complete test suite.

Following options are supported by the test suite:

Option	Description
<code>--fastMode</code>	Runs the test suite with a reduced amount of data
<code>--backintVersion</code>	Sets the backint protocol version used by the certification suite. Possible options are "1.5" and "1.0". Default value is "1.5"

Cleanup data created by test suite execution:

Notice that in case of a successful execution there is no need to clean up the test environment, because at the end the entire database installation inclusive test environment will be removed. The generated data is contained in the database schema and can be deleted using the following SQL command "DROP SCHEMA BNR_DATA CASCADE"

3.3 Test Catalog

3.3.1 Testcase #1: Recover Data

- Create data
- Create backup with "prefix_1"
- Recover backup with "prefix_1"

3.3.2 Testcase #2: Recover Database (point in time recovery)

- Create data
- Create backup with "prefix_2"
- Create data
- Recover up to the latest point in time
- Remember this point in time
- Delete data
- Create backup with "prefix_2"
- Create data
- Recover up to the latest point in time
- Recover until point in time from step 5

3.3.3 Testcase #3: Inquire with wildcard #NULL

- Call backint inquire function with #NULL and data backup parameter file.
- Check that the returned backups match the backup catalog

3.3.4 Testcase #4: Inquire single backup

- Inquire the backup made in testcase #1

3.3.5 Testcase #5: Inquire oldest two backups

- Inquire the backup made in testcase #1 and the first backup made in testcase #2

3.3.6 Testcase #6: Inquire younger backup

- Inquire the backups made in testcase #2 for all volumes using single backint calls with specified filename and #NULL as wildcard

3.3.7 Testcase #7: Inquire youngest and oldest backup

- Inquire the first and the second backup made in testcase #2 using one backint call and fully specified backups

3.3.8 Testcase #8: Inquire youngest backup single ebid

- Execute a single inquire call for each volume from the youngest data backup of testcase #2

3.3.9 Testcase #9: Delete youngest backup

- Delete the second backup made in testcase #2

3.3.10 Testcase #10: Inquire oldest backup single ebid

- Inquire all parts of the first backup from testcase #2 using single backint calls

3.3.11 Testcase #11: Inquire deleted backup

- Inquire all parts of the deleted second backup from testcase #2 using one backint call
- Confirm that the inquired backup is not found

3.3.12 Testcase #12: Inquire creation timestamp

This test case is only valid for backint 1.5

- Inquire the backup made in testcase #1
- Confirm that the creation time returned by backint is between UTC_START_TIME and UTC_END_TIME reported by M_BACKUP_CATALOG

3.4 Test Report

The result of the certifications suite is written on stdout only. In order to store the result one can use the linux program tee as below:

```
python test_backup_backint_certification.py 2>&1 | tee certification.log
```

Additional information is stored in a subdirectory of the trace directory of the SAP HANA database. It is located in \$DIR_INSTANCE/<hostname>/trace and has the name backint_certification_YYYYMMDD-HHMMSS (e.g. /usr/sap/MBY/HDB01/lu349879/trace/backint_certification_20170727-135012)